Pioneer Junior College Preliminary Examinations 2008 H2 Biology

Paper 1

1	**D**	2	**B**	3	**B**
4	**B**	5	**A**	6	**C**
7	**B**	8	**A**	9	**C**
10	**D**	11	**A**	12	**B**
13	**C**	14	**C**	15	**D**
16	**D**	17	**C**	18	**C**
19	**A**	20	**D**	21	**C**
22	**C**	23	**A**	24	**B**
25	**C**	26	**D**	27	**B**
28	**D**	29	**C**	30	**C**
31	**C**	32	**C**	33	**D**
34	**A**	35	**C**	36	**D**
37	**A**	38	**C**	39	**D**
40	**D**				

Paper 2
Question 1
The diagram below shows nuclear division occurring in an animal cell.

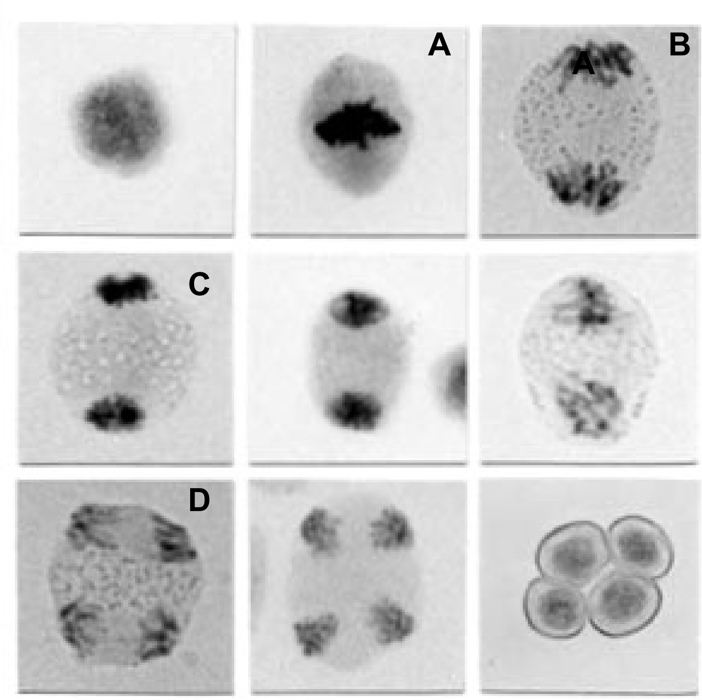


Fig. 1.1

a)
i) Identify stages A to D. [2]

A Metaphase I

B Anaphase I / early telophase I

C Telophase I

D Anaphase II

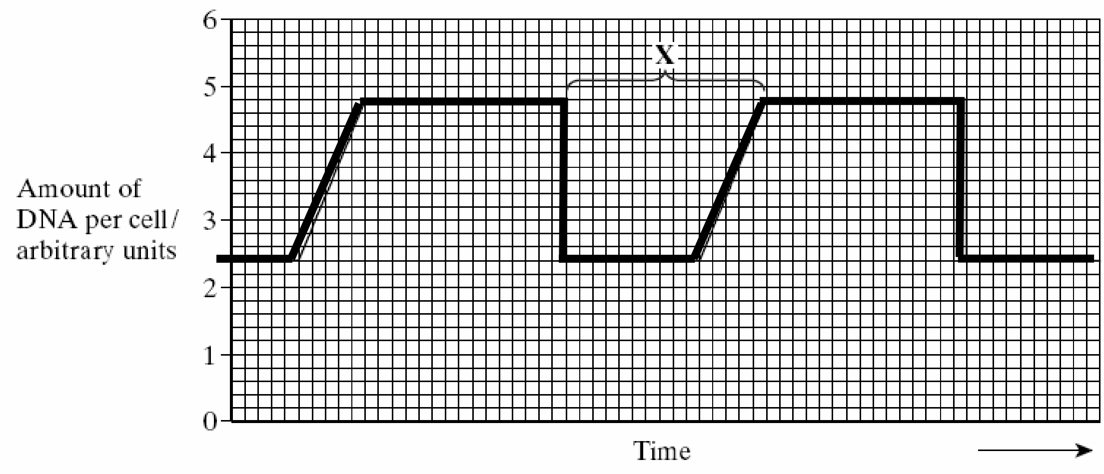
ii) State the type of nuclear division shown in Fig. 1.1 [1]

Meiosis

iii) Explain your answer to part (ii). [1]

2 nuclear divisions;
4 nuclei/cells formed;

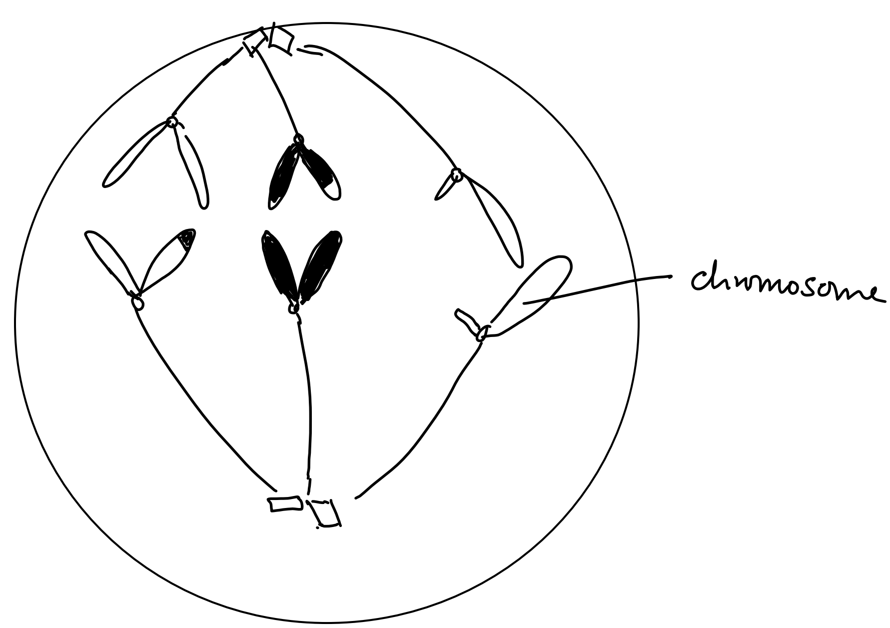
b) The graph below shows the relative amounts of DNA per cell during two successive cell divisions in a plant.



i) Describe the events occurring at X. [3]

Cytokinesis occurred;
amount of DNA in the cell dropped from 4.8 units to 2.4 units;
Plateau in graph;
Cell growth;
Interphase occurs after cell growth;
Doubling of amount of DNA
Graph shows increase/ double from 2.4 units to 4.8 units

c) Draw the behaviour of chromosomes (2n = 6) undergoing anaphase II during nuclear division in the cell given below. [3]



[total: 10]

Question 2

a) Protein synthesis is necessary for the survival of cells where important proteins are synthesised. Protein synthesis involves both the process of transcription and translation. Any disruption to these processes may cause the cell to be dysfunctional.

i) Describe the importance of the DNA-binding domain and protein-binding domain of transcription factors. [2]

DNA-binding domain recognizes and;
binds to specific DNA sequences;
Protein-binding domain contains binding sites;
For other transcription factors or proteins (eg RNA polymerase) to bind to;
To form transcription initiation complexes;
importance – initiation / regulation of transcription;

ii) A retrotransposon (a 'jumping gene' containing non-coding DNA) was accidentally inserted at the TATA region in the promoter of a gene. Explain how this would affect transcription of that gene. [2]

change in sequence of bases at TATA region;
DNA binding domain of transcription factor cannot recognize the TATA region;
unable to form hydrogen bonds;
other transcription factors and RNA polymerase cannot bind to promoter;
transcription is unable to occur;

iii) State precisely where in the cell translation occurs. [1]

Ribosomes;
in cytoplasm/ RER;
Reject: cytoplasm only

iv) Describe how the structure of the ribosome is adapted for translation. [3]

Large subunit contains <u>peptidyl transferase</u>;
for formation of <u>peptide bonds</u> in between amino acids;
Holds the translation initiation complex in place;
contains Peptidyl, aminoacyl and exit sites;
Peptidyl site contains the tRNA with the polypeptide chain attached;
Aminoacyl site is for binding of the new aminoacyl tRNA complex;
Exit site for tRNA to be released from ribosome;
Small subunit for attachment of mRNA;

v) A mutation was found in a gene that codes for the capping enzyme complex. Explain how this mutation would affect translation. [2]

capping enzyme complex is non-functional;

EITHER
capping enzyme complex cannot synthesise 5' cap for pre-mRNA;
mRNA cannot exit the nucleus to ribosomes;
for translation;

OR

incorrect 5' cap formed;
mRNA cannot bind to small ribosomal subunit;
translation is unable to occur;

[total: 10]

Question 3

a) Describe 2 differences between the structure of prokaryotic and eukaryotic genomes. [2]

Prokaryotic genome	Eukaryotic genome
1 origin of replication	**Multiple origins of replication**
absence of telomeres	**Presence of telomeres at ends of chromosomes**
Circular/closed loop	**Linear**
Absence of introns	**Presence of introns**
Found in nuceloid region/genome in contact with cytoplasm	**Found in nucleus**
Absence of centromeres	**Presence of centromeres**
Non-histone proteins associated with DNA	**Histone proteins associated with DNA to form nucleosome**

Figure 3.1 below represents the processing of pre-messenger RNA.

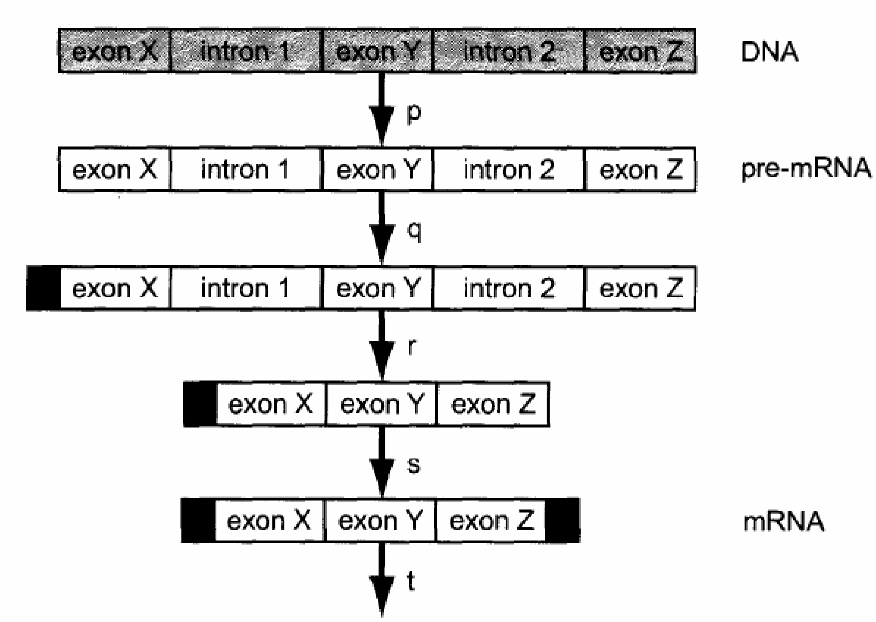


Fig. 3.1

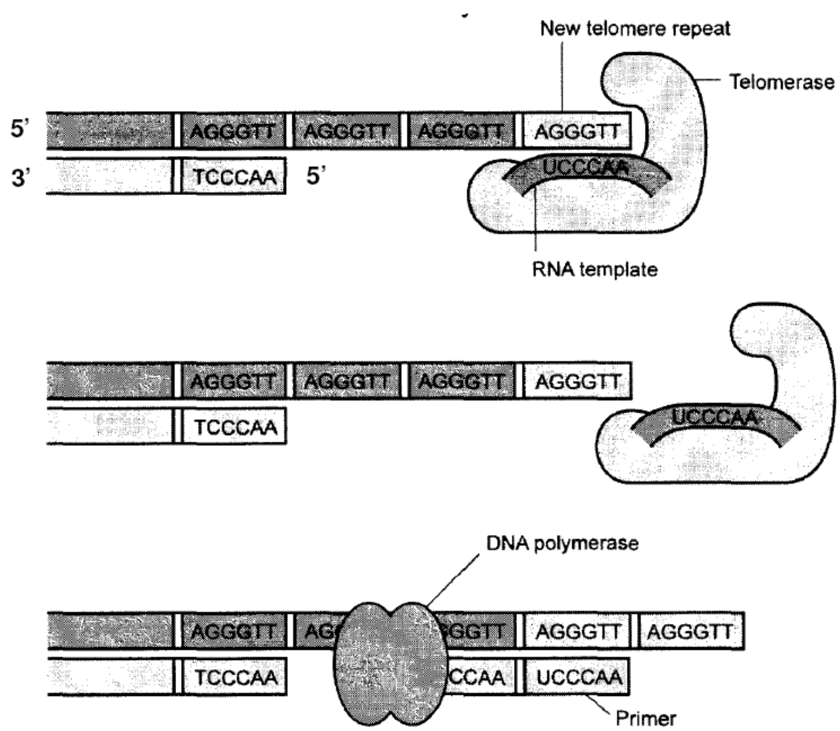
b) Describe the events **r** to **t**. [5]

r: **splicing process;**
snRNP (small nuclear ribonucleoproteins) recognize splice sites;
spliceosomes interact with splice site to remove introns 1 and 2;
and joins exons X, Y and Z together;

s: **addition of <u>poly adenylate (A) tail/ polyadenylation;</u>**
which is a string of adenosine residues;
using enzyme poly (A) polymerase;

t: **functional/mature mRNA forms;**
Leaves the nucleus via <u>nuclear pore</u>;
To undergo <u>translation</u>;
At the <u>ribosomes</u> in the cytoplasm;

c) Fig 3.2 shows the action of telomerase enzyme.



i) State whether the new telomere repeat is added to the daughter or parent DNA strand. Explain your answer. [2]

Parent DNA strand
Parental DNA ends with 3' end where elongation can occur;
New telomere must be added to parental strand to enable replication of telomere on the daughter strand;

ii) Explain why telomerase activity is essential in some unicellular organisms. [1]

cell division is their mode of reproduction;
division cannot be limited;

[total: 10 marks]

Question 4

a) Pure-breeding fruit flies, *Drosophila melanogaster*, with wild-type bodies and vestigial wings were crossed with pure-breeding flies with black bodies and wild-type wings. The F1 all had wild-type bodies and wild-type wings.
A test cross was then carried out with F1.

i) State the ratio of phenotypes expected in a dihybrid cross such as this test cross. [1]

1 Wild-type body, vestigial wings
1 Wild-type body, wild type wings
1 Black body, vestigial wings
1 Black body, wild-type wings

The actual results from this cross were as follows:

Wild-type body, vestigial wings	1252
Wild-type body, wild type wings	254
Black body, vestigial wings	248
Black body, wild-type wings	1246

ii) Draw a genetic diagram to show how this results. [4]

Use symbols: R – wild-type body r – black body
T – wild-type wings t – vestigial wings

Parental phenotype: wild-type bodies, wild-type wings (F1) X Black body, vestigial wings

Parental genotype Rt/rT X rt/rt

Gametes: Rt rT (parental types) X rt

RT rt (recombinants)

gametes	**Rt**	**rT**	**RT**	**rt**
rt	**Rt/rt wild-type body vestigial wings**	**rT/rt black body wild-type wings**	**RT/rt wild-type body wild-type wings**	**rt/rt black body vestigial wings**
	Parental types	**Parental types**	**Recombinant**	**Recombinants**
	1252	**1246**	**254**	**248**

iii) Explain the discrepancy between the expected and actual results. [4]

1. genes for body type and wing shape are <u>linked/found on the same chromosome</u>;
2. which means they are usually carried together during gamete formation;
3. alleles for wild type body (R) and vestigial wing (t) are found on one chromosome;
4. while alleles for black body (R) and wild-type wing (T) is found on another chromosome;
5. during gamete formation from F1, there will be a higher proportion of gametes carrying the parental types – Rt and rT;
<u>6. crossing over between the 2 alleles/genes on non-sister chromatids</u> during prophase 1 of meiosis can also occur ;
7. to form recombinant gametes – RT and rt;
8.since <u>crossing over is a chance event</u>, there are lower chances of getting recombinant gametes;
9. resulting is a smaller proportion of offspring having recombinant phenotypes;

b) In another experiment, another phenotype, eye colour, was examined together with wing shape. Pure-breeding *Drosophila melanogaster* with red eyes and vestigial wings were crossed with pure-breeding flies with black eyes and wild-type wings. The F1 all had black eyes and wild type wings.
Females of F1 were then crossed with males of F1.

The experimental results of the F2 generation are as below:

F2 generation

Black eyes and wild-type wings	510
Black eyes and vestigial wings	173
Red eyes and wild-type wings	165
Red eyes and vestigial wings	52

i) Use the chi-square (X^2) test and the table of probability shown in table 4.1 to find the probability of the results of the F2 generation departing significantly from expectation.
Show your working.

$$\chi^2 = \Sigma \frac{(O-E)^2}{E} \qquad v = c - 1$$

where Σ = 'sum of…' O = observed 'value'
v = degrees of freedom E = expected 'value'
c = number of classes

Distribution of X^2

degrees of freedom	probability, p				
	0.10	*0.05*	*0.02*	*0.01*	*0.001*
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Degree of freedom = 4-1 =3

Category	O	E	O-E	$(O\text{-}E)^2$	$\frac{(O\text{-}E)^2}{E}$
Black eyes and wild-type wings	**510**	**506.25**	**3.75**	**14.0625**	**0.0278**
Black eyes and vestigial wings	**173**	**168.75**	**-4.25**	**18.0625**	**0.107**
Red eyes and wild-type wings	**165**	**168.75**	**3.75**	**14.0625**	**0.083**
Red eyes and vestigial wings	**52**	**56.25**	**4.25**	**18.0625**	**0.321**

X^2 = **0.5388**

Probability = **less than 10% that the results depart significantly from expected**
Or **more than 10% that the difference between observed and expected is due to chance**
[4]

[total: 13]

<u>Question 5</u> [7 m]

The pathway of carbon during the reactions of photosynthesis was discovered using the unicellular green algae Chlorella. The Chlorella culture was supplied with $^{14}CO_2$ and illuminated with varying length of time. Some algae were removed at intervals and immediately killed by boiling in alcohol. This provided an extract which was examined using chromatographic techniques and any radioactive compounds present showed up on X-ray film as blackened areas.

After 15 seconds of illumination, the extract produced three prominent blackened areas which were due to the presence of three compounds:

i. glycerate phosphate (GP/ PGA)
ii. ribulose bisphosphate (RuBP)
iii. triose phosphate (TP/ PGAL)

a) In a second series of experiments the algae were illuminated in the presence of $^{14}CO_2$ for 15 seconds. Illumination of the algae continues for a further 1 minute after all $^{14}CO_2$ present had been removed. Some algae were then removed, killed and the extract examined. A distinct blackened area was identified on the X-ray film.

Name the compound that was found in this blacken area. Explain your answer. [2]

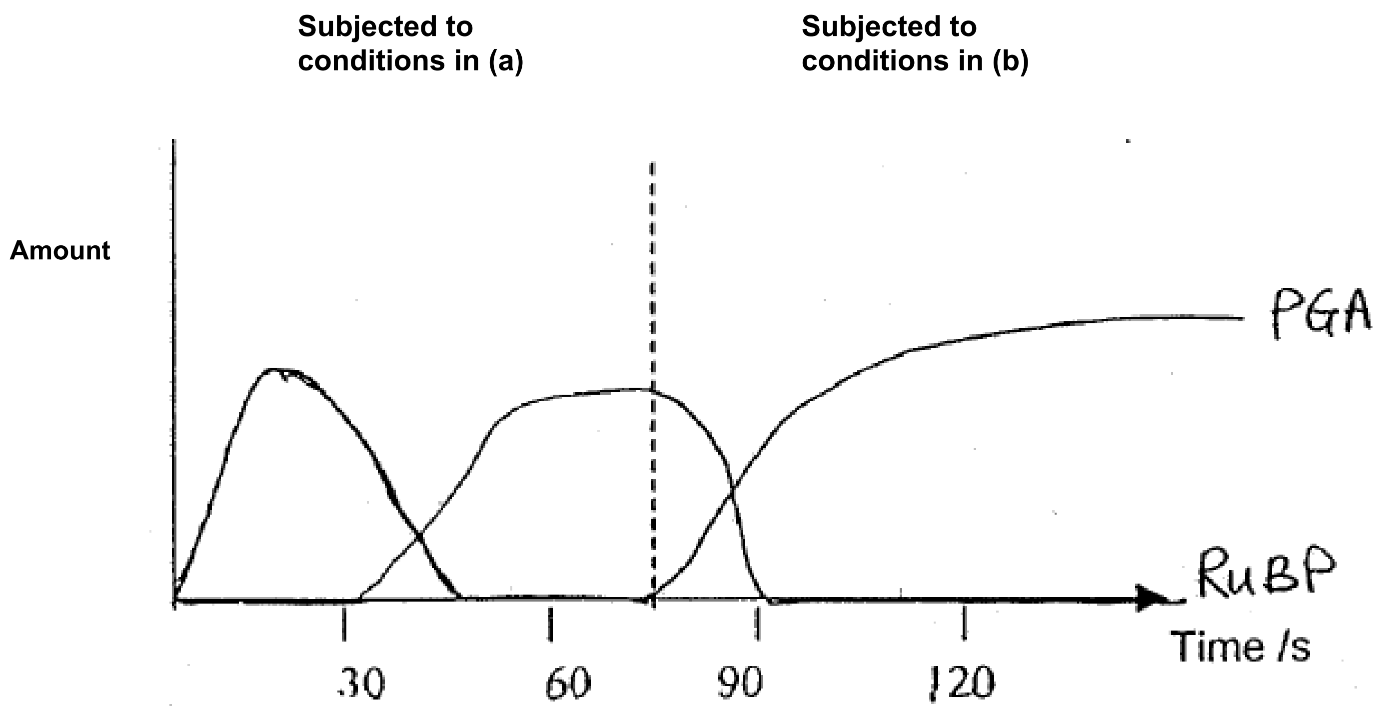
- **<u>Ribulose bisphosphate (RuBP);</u>**
- **In the absence of CO_2, RuBP is not utilised in carboxylation to form glycerate phosphate (GP/ PGA);**
- **However, in the presence of light, light reaction allows the formation of ATP and NADPH;**
- **Labelled PGA/GP can be reduced to triose phosphate (TP)/G3P;**
- **which in turn is used in the regeneration of RuBP, thus, accumulation of RuBP;**

b) In another series of experiments, the algae were illuminated in the presence of $^{14}CO_2$ for 15 seconds and then placed in the dark for 1 min in the presence of $^{14}CO_2$. Some algae were removed, killed and the extract examined. A distinct blackened area was identified on the X-ray film.

Name the compound that was found in this blackened area. Explain your answer. [2]

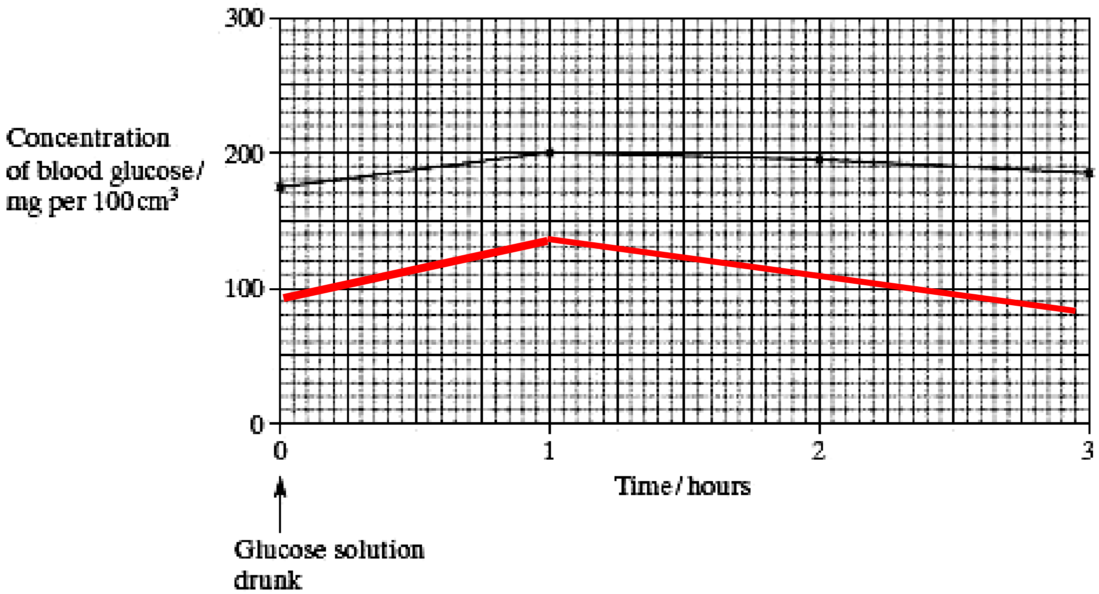
- **Glycerate phosphate (GP/ PGA);**
- **CO2 combines with RuBP to form GP/PGA in the absence of light;**
- **However, in the absence of light, light reaction cannot proceed, thus, ATP and NADPH cannot be made;**
- **GP/PGA cannot be reduced to TP/G3P**
- **and RuBP cannot be regenerated;**

c) On the graph below, show how the amounts of the labelled compounds named in (a) and (b) will change if the algae is *first* subjected to the conditions in (a) *then* to conditions in (b). [3]



Question 6 [10 m]

Type 1 diabetes is often due to the inability to produce insulin. In an investigation, a man with type 1 diabetes drank a glucose solution. The concentration of glucose in his blood was measured at regular intervals. The results are shown in the graph below.



a) Suggest **two** reasons why the concentration of glucose decreased after 1 hour even though this man's blood contained no insulin. [2]

- **glucose is used as substrate in cell respiration to provide energy for metabolic activities;**
- **glucose is excreted in urine;**

b) The investigation was repeated on a man who did not have diabetes. The concentration of glucose in his blood before drinking the glucose solution was 80 mg per 100 cm^3. Sketch a curve on the graph above to show the results you would expect. [1]

- **curve starting from 80 mg 100 cm^{-3} increasing to about 120 mg 100 cm^{-3} and decreasing back to 90 mg 100 cm^{-3}**

c) Explain how the blood insulin is homeostatically regulated in a normal individual. [3]

- **hyperglycaemia/increased blood glucose concentration above set point;**
- **stimulates pancreatic β-cells of islets of langerhans to secrete insulin;**
- **insulin promotes glucose uptake; and utilisation in target cells; increases glycogenesis in liver cells; increases protein synthesis; stimulates lipogenesis; inhibits glycogenolysis; and gluconeogenesis; (max 1 m for effects of insulin)**
- **consequent decrease in blood glucose concentration back to set point;**
- **exerts negative feedback on pancreatic β-cells to inhibit insulin secretion;**
- **causing fall in blood insulin level**

Figure 6.2 shows the normal sequence of cell signalling events when insulin reaches its target cell and binds to an insulin receptor resulting in an increase in the number of glucose channels in the cell surface membrane.

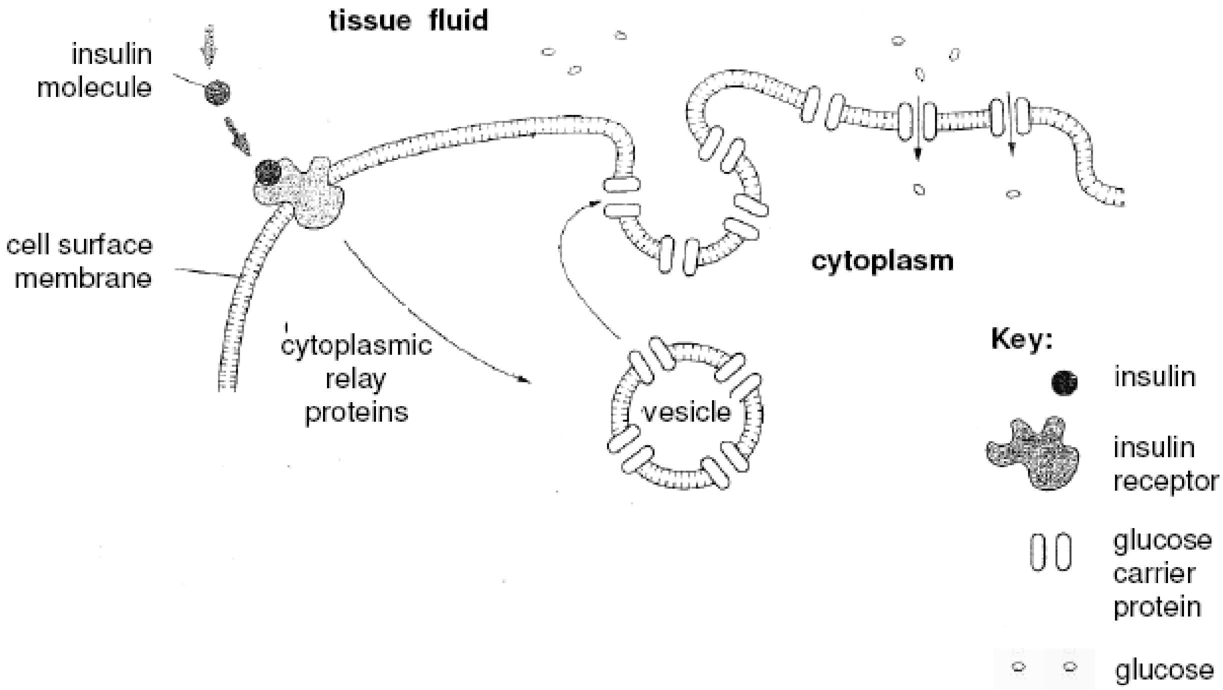


Fig. 6.2

d) With reference to Fig 6.2 and using your biological knowledge, describe the events that ultimately result in an increase in glucose uptake. [4]

- **binding of insulin causes the <u>2 receptor sub-units/polypeptide</u> chains to form <u>a dimer</u>;**
- **dimerisation <u>activates the tyrosine kinase region</u> of each sub-unit;**
- **leading to <u>phosphorylation of tyrosine residues</u> on the tail of subunits;**
- **specific <u>relay proteins</u> in cell bind to phosphorylated tyrosines and become activated;**
- **activated relay proteins trigger <u>transduction pathway/phosphorylation cascade</u>;**
- **protein <u>kinases are activated</u>;**
- **causing <u>vesicles with glucose carrier proteins</u> to move to cell surface membrane and <u>fuse</u> with it;**
- **glucose carrier proteins are then inserted in the membrane as <u>glucose channels</u>;**
- **increase in no of glucose transporter proteins results in <u>increase in glucose uptake</u> by cells**

Question 7

Figure 7.1 below shows part of two nerve fibres and a synapse. The figures indicate the value in mV of the potential across the membrane between the cytoplasm of the fibres and the extracellular fluid at intervals along each fibre.

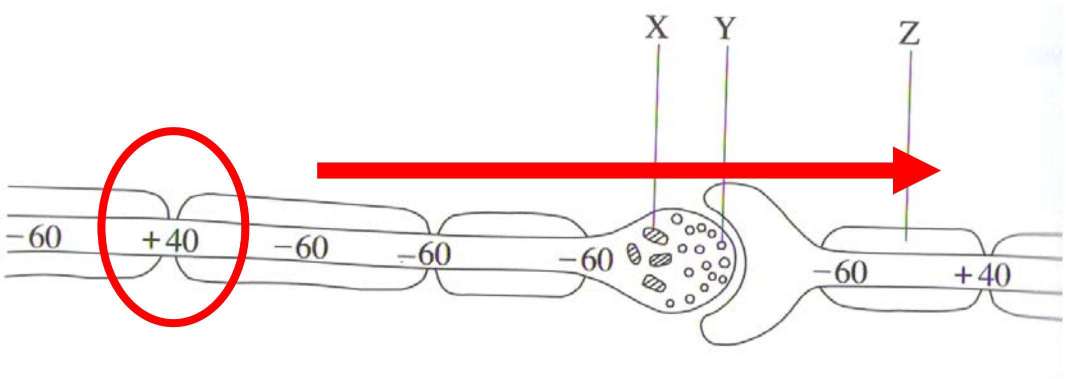


Fig. 7.1

ai) Draw a circle around one region of the diagram where an action potential exists. Explain your choice. [2]

correct circle – 1 mark
the neuron is depolarized here with a value of +40 mV
this shows influx of Na+

ii) By means of an arrow on the diagram, indicate the direction in which action potentials would normally travel along these fibres. Explain why the transmission of impulse is uni-directional across two nerve cells. [2]

correct arrow – 1 mark
neurotransmitter and its vesicles are found only in the pre-synaptic knob;
receptors for neurotransmitter is found only in the post-synaptic membrane;

iii) Identify structures X and Y, and explain how each is involved in the transmission of nerve impulse. [1]

Structure X: **mitochondrion**;
Explanation: **Production of energy in the form of ATP;**
For active processes such as exocytosis;

Structure Y: **neurotransmitter vesicles/ acetylcholine vesicles/synaptic vesicles;** [2]

Explanation:
1. **vesicles will <u>fuse with the presynaptic membrane</u> when an action potential reached the pre-synaptic knob;**
2. Release neurotransmitter/ acetylcholine into the <u>synaptic cleft</u>;
3. Diffusion of neurotransmitter across synaptic cleft to reach receptor at post-synaptic membrane;
4. opening of specific/ Na channel, leads to EPSP;

b) Figure 7.2 shows a normal nerve-muscle junction.

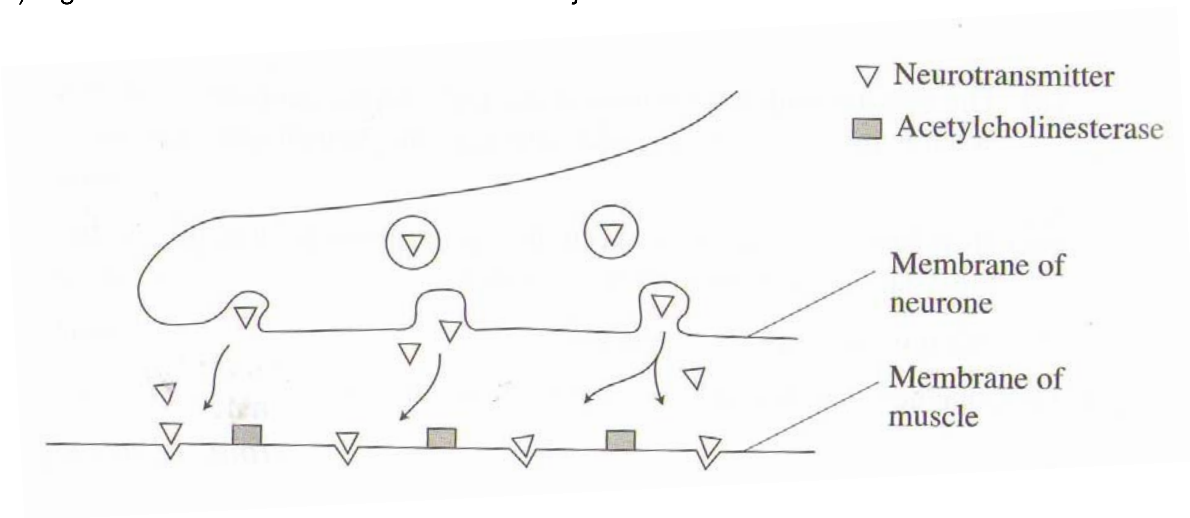


Fig. 7.2a

Using the evidence from Fig. 7.2a and Fig. 7.2b, briefly describe and explain the effect on muscle activity of i) organophosphates and ii) botulin toxin. [3]

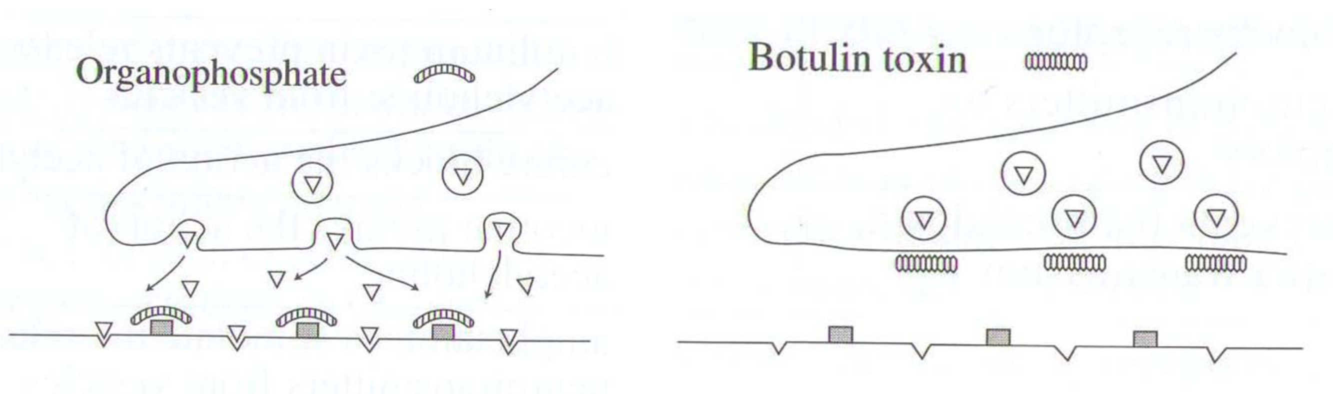


Fig. 7.2b

i) organophosphate:

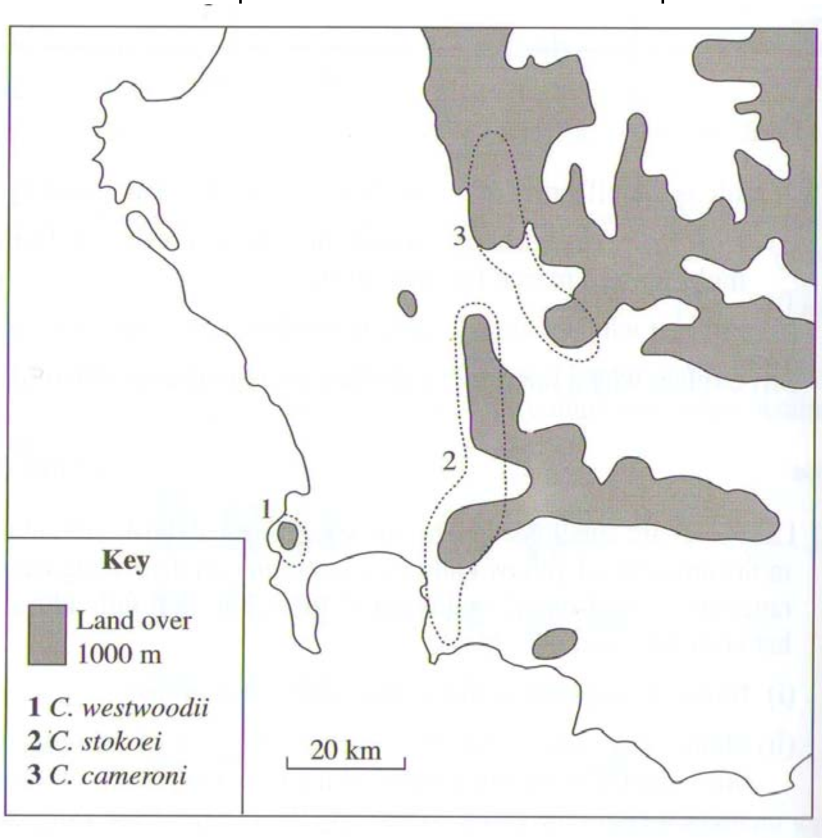
it is an inhibitor of acetylcholinesterase;
prevent enzyme from breaking down acetylcholine;
muscle activity is constantly stimulated;

ii) botulin toxin

prevents pre-synaptic vesicles from fusing with membrane;
thus prevent acetylcholine from being released into the synaptic cleft;
muscle cannot be stimulated;

Question 8

The beetles being to the genus *Colophon* are unable to fly and are found on hilltops in South Africa. The dotted lines on the map show the distribution of three species.



a) Suggest an evolutionary explanation for each of the following statements.

i) All of these beetles are **similar** general appearance. [1]

they have common ancestor;

ii) There are **differences** between the species of Colophon found in the three areas. [3]

The different environments act as different <u>selection pressures</u>;
Variations exist within populations;
Those which adapt favourably to different environments;
Survives and pass on favorable genes to offspring;
Results in divergent evolution;
The fact that they are unable to fly / geographical isolation;
They are unable to interbreed/ reproductive isolation;
Over time, each will evolve into a new species;

b) Many types of evidence are now used to trace the evolutionary history of various species. Biochemical evidence can be used to determine the degree of similarity between organisms. In this technique, the similarities between related proteins can be determined.
The following table compares the number of different amino acids in the cytochrome *c* molecule from a variety of species.

	Human	Monkey	Pig	Horse	Dog	Rabbit	Kangaroo	Chicken	Duck	Rattlesnake	Turtle	Tuna	Moth	Neurospora	Candida	Yeast
Human	0															
Monkey	1	0														
Pig	10	9	0													
Horse	12	11	3	0												
Dog	11	10	3	6	0											
Rabbit	9	8	4	6	5	0										
Kangaroo	10	11	6	7	7	6	0									
Chicken	13	12	9	11	10	8	12	0								
Duck	11	10	8	10	8	6	10	3	0							
Rattlesnake	14	15	20	22	21	18	21	19	17	0						
Turtle	15	14	9	11	9	9	11	8	7	22	0					
Tunafish	21	21	17	19	18	17	18	17	17	26	18	0				
Moth	31	30	27	29	25	26	28	28	27	31	28	32	0			
Neurospora	48	47	46	46	46	46	49	47	46	47	49	48	47	0		
Candida	21	21	20	21	49	20	21	21	21	21	23	48	47	42	0	
Yeast	45	45	45	46	45	45	46	46	46	47	49	47	47	41	27	0

i) Explain which species diverge most recently from the turtle. [2]

duck;
the number of different amino acids is 7;
which is the least;

ii) It is suggested that using changes in amino acid sequences could not be an accurate representation of the genetic similarity between two species. Explain why this is so. [2]

this could be due to degenerate property of triplet code;
a change in the last nucleotide does not change the type of amino acid;
this is also known as silent mutation;
thus even if there is no change in amino acid sequence, it does not necessary mean that there is no change to the DNA sequence;
amino acid sequence only reflect coding region of DNA sequence;
changes in non-coding region will not be reflected;

iii) Explain why cytochrome *c* molecule is present in **all** the listed species. [2]

cyctochrome *c* is present in the earliest species/common ancestor;
and because it is an important molecule for aerobic organisms;
it should persist to the most recent species/to all species;
despite some changes to the amino acid sequences;

9a) With reference to a named virus and its reproductive cycle, outline how specialized transduction could result in a stable transfer of genes into a recipient bacterium. [12]

1. **Definition: Specialized transduction is a form of DNA transfer using phages (the viruses that infect bacteria) to carry bacterial genes from one host cell to another;**
2. **lambda phage;**
3. **adsorption – attachment of tail fibre to host cell;**
4. **penetration – tail sheath contracts to inject viral DNA into host cell;**
5. **forms a prophage by integrating viral DNA into the host cell chromosome (at insertion sites);**
6. **in response to environmental stimuli induction may occur;**
7. **switches from lysogenic to lytic cycle;**
8. **whereby prophage excised from/ exits from host chromosome;**
9. **host / bacterial genes <u>adjacent</u> to either side of prophage insertion sites may be excised together with prophage;**
10. **and then packaged into a capsid;**
11. **host cell is lysed and new phages released;**
12. **new phage injects its <u>hybrid/recombinant</u> DNA into recipient bacterial cell;**
13. **homologous recombination may occur/ newly introduced bacteria genes replaces <u>homologous</u> genes on the recipient chromosome;**
14. **resulting in transfer of donor genes into recipient bacterium;**

b) Explain how viral infections could cause disease in animals. [5]

1. **cause phenotypic / degenerative changes in host cell leading to altered functions;**
2. **inhibits normal DNA, RNA or protein synthesis in host cell leading to altered cell functions;**
3. **viral proteins and glycoproteins change <u>antigenic surface</u> of host cell's cytoplasmic membrane, resulting in it being recognized as foreign and destroyed by the host's immune system;**
4. **deplete the host cell of <u>cellular materials</u> eg amino acids, nucleotides that are essential for normal functioning;**
5. **trigger release of hydrolytic enzymes from host cell lysosomes leading to cytolysis of infected host cells;**
6. **Inhibits expression of host genes by inserting viral genome into host DNA / chromosome;**
7. **Some viruses are oncogenic causing normal cells to become malignant (cell transformation by <u>oncogenic</u>/cancer-causing viruses);**
8. **viral genome may be expressed in the host to produce toxins that damage/destroy host cells;**

c) Hepatitis A is a disease caused by a virus which can permanently damage the liver and pancreas. Suggest why the virus causes damage only to some types of human cells. [3]

1. **only susceptible host cells have specific receptors on their cell surface membrane;**
2. **that allow for complementary binding between antireceptors on the virus and surface receptors on host cell;**
3. **In the absence of its specific receptor, the virus cannot adsorb and penetrate into host cell and hence cannot infect**

10a) Explain how natural selection may bring about evolution. [7]

1. **Obs 1: Populations have great reproductive potential/ reproduce more than needed to replace themselves;**
2. **Obs 2: Numbers remain approximately constant;**
3. **Ded 1: Many fail to survive/ there is a struggle for existence;**
4. **Being checked by environmental factors as pressure;**
5. **eg 1: Competition for resources;**
6. **eg 2: Predation/ diseases;**
7. **Obs 3: Variation in the population;**
8. **Ded 2: Individuals with genetic variations best adapted to the environment;**
9. **were more likely to survive and;**
10. **reproduce/pass on their genes to the next generation;**
11. **Selection of such phenotypes would lead to a change in the frequency of particular alleles and genotypes;**
12. **proportion of population showing favorable traits will increase;**
13. **proportion of population showing unfavorable traits will decrease;**
14. **Over time it would lead to evolutionary change;**
15. **New species arising from existing ones;**

(b) Describe the mechanisms that can cause changes in allele frequencies in a population. [6]

Mutation

1. **Only mutation produces new alleles**
2. **can immediately change the gene pool of a population by substituting one allele for another.**

Non-random breeding

3. **Inbreeding usually occurs between closely related partners.**
4. **mating also usually occurs between organism in closer proximity;**
5. **sexual selection ensures that certain individuals within the population have an increased reproductive potential;**
6. **so their alleles are more likely to be passed on to the next generation;**

Genetic drift

7. **Refers to the fact that variation in gene frequencies within populations can occur by chance events rather than by natural selection.**
8. **In a small population not all the alleles which are representative of that species may be present/ Some alleles may be absent and others may be disproportionately represented.**

Founder effect

9. **Arises when a small population becomes split off from the parent population**

Bottleneck effect

10. **Arises when a population is drastically reduced in size by sudden changes in the environment such as natural disasters or human interference ;**

Gene flow

11. **Refers to movement of alleles from one population to another due to migration;**
12. **resulting from the movement of fertile individuals or their gametes;**

Natural selection

13. **Differential success in reproduction**
14. **Results in alleles being passed along to the next generation in numbers disproportionate to their relative frequencies in the present generation.**

Explain, with appropriate examples, how recessive alleles may be preserved in a natural population. [7]

1. **Diploidy – having 2 sets of chromosomes;**
2. **A large proportion of recessive alleles in a population exist in carrier heterozygotes;**
3. **therefore the recessive allele is not expressed, thus is protected from natural selection;**
4. **this is known as <u>heterozygote protection</u>;**

5. **Balancing selection - occurs when natural selection maintains two or more forms in a population;**

6. **Some individuals who are heterozygote at a particular locus have greater fitness than do both kinds of homozygotes;**
7. **in sickle cell anemia, homozygous recessive individuals usually die before reaching adulthood thereby eliminating two recessive alleles from the populations;**
8. **Heterozygote individuals possessing a single sickle-cell allele have increased resistance to malaria;**
9. **Homozygous dominant individuals usually die from malaria;**
10. **heterozygote advantage/ Selective advantage of heterozygotes;**
11. **Leading to higher frequency of sickle-cell allele in malaria-affected countries;**

12. **Frequency-Dependent Selection - <u>fitness of a phenotype declines if it becomes too common in the population;</u>**
13. **Example: Scale-eating fish (*Perissodus microlepis*) of Tanganyika in Africa;**
14. **Right-mouthed allele is dominant to the left-mouthed allele;**
15. **Prey species guard against attack from whatever phenotype of scale-eating fish that is most common in the lake;**
16. **selection favours whichever mouth phenotype is least common;**
17. **(balancing selection) keeps the frequency of each phenotype close to 50%;**

Paper 3
Question 1

a) The gene map of a plasmid, which has been used for genetic engineering, is shown below in Fig. 1.1. It contains two antibiotic resistance genes. The positions of several restriction endonuclease binding sites are also shown below.

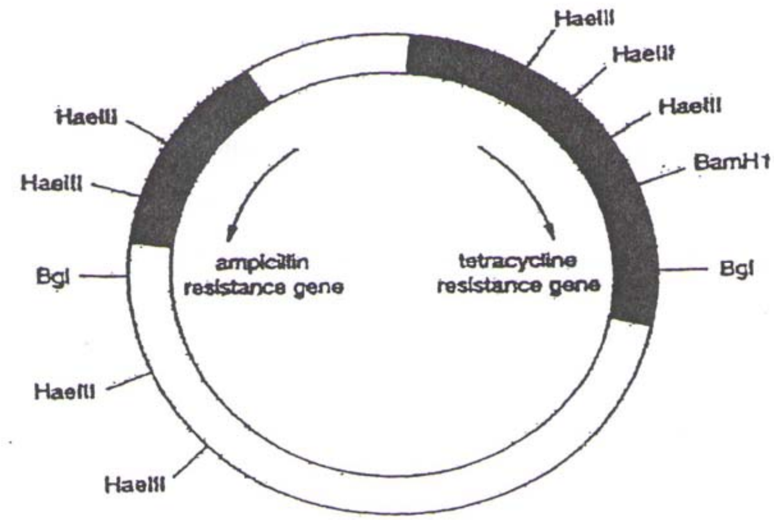


Fig 1.1

i) Describe the function of restriction endonuclease. [1]

Recognises and binds to restriction/specific sites;
Catalyses the breaking of <u>phosphodiester bonds</u> between specific base pairs;

ii) State which enzyme(s) would be the most suitable for the cleavage of the plasmid and the DNA coding for gene of interest? Explain briefly. [1]

Bam H1;
Only cleaves at one location/genetic marker;
This allow for selection;
(or the converse – too many RE sites, or do not cut at the marker)

b) The DNA containing the gene coding for insulin and plasmid with genes coding for resistance against tetracycline, ampicillin and streptomycin were mixed together with the appropriate restriction enzyme. A heat shock treatment was applied to bacteria to allow the plasmids to enter the cells. The bacteria was plated and grown in the following media as shown in Fig. 1.2.

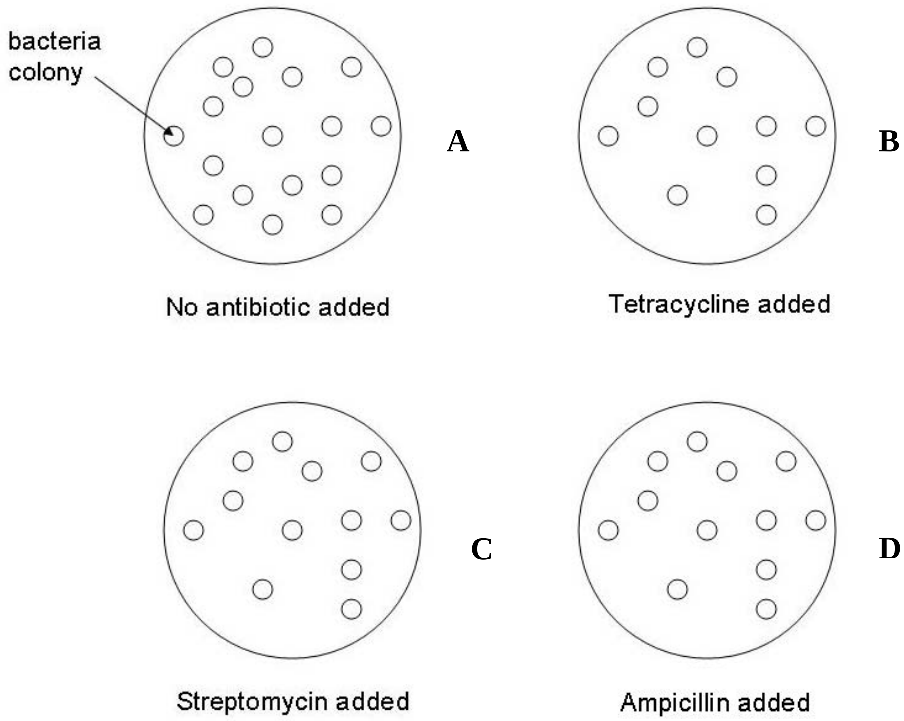


Fig. 1.2

i) How is the gene for human insulin **usually** obtained? Describe the process.

Obtain the mRNA from pancreatic cells/ β cells of the islets of Langerhans;
Form cDNA from mRNA;
Using reverse transcriptase;
DNA polymerase is used to form the complementary strand of cDNA;

ii) Explain the advantage of using the process described in part (i). [1]

No introns in cDNA;
Excising of introns from mRNA is not needed as bacteria/host cells do not having splicing machinery;
Direct synthesis of pro-insulin;

iii) Explain why 2 marker genes are needed for identification of transformed cells. [2]

1 marker gene is used to identify transformed cells/ cells that picked up a plasmid;
Another marker gene is used to identify cells with recombinant DNA;

iv) With reference to Fig. 1.2 above, state which plate contains the gene coding for insulin inserted. [1]

A, C or D

v) Explain your answer to part (iv). [2]

same number of colonies are present on plates containing ampicilin and streptomycin;
both genetic markers are intact, colonies are transformed with plasmids;
Cells growing on plate with tetracycline has 1 less colony;
That missing colony has no tetracycline resistance;
Gene coding for resistance has been inactivated by insertion of insulin gene;

vi) Can the insulin obtained directly from the bacteria be used in humans? Why? [2]

No;
Insulin obtained directly from bacteria is in the form of pro-insulin;
Extra C chain;
Enzymes needed to cut away the C chain and join the A and B chains to form insulin;

c) Distinguish between a genomic and cDNA library. [3]

Genomic library	**cDNA library**
Contains all coding and non-coding sequences	**Expressed genes in the tissue – lack non-coding regions/ only coding sequences**
Cell's DNA cut using RE and inserted into vectors	**mRNA → cDNA using reverse transcriptase which is then inserted into vectors**
Need vectors ranging from plasmids to BAC to accommodate larger DNA sequences	**cDNA is relatively short, plasmids and λ phage often used as vectors**
A single gene can be spread out over different clones - A lot of DNA to isolate and analyze	**only contain coding sequences for that gene, which reduces the amount of DNA one has to isolate and analyze**
To study control of gene expression	**Isolate a particular gene and analyse its properties/ functions**

[total: 15]

Question 2

A length of DNA from one of a pair of homologous chromosomes is shown in Fig. 2.1. The target sites of EcoR1 are shown by arrows and the length of DNA between the target sites is given in kilobases (kb).

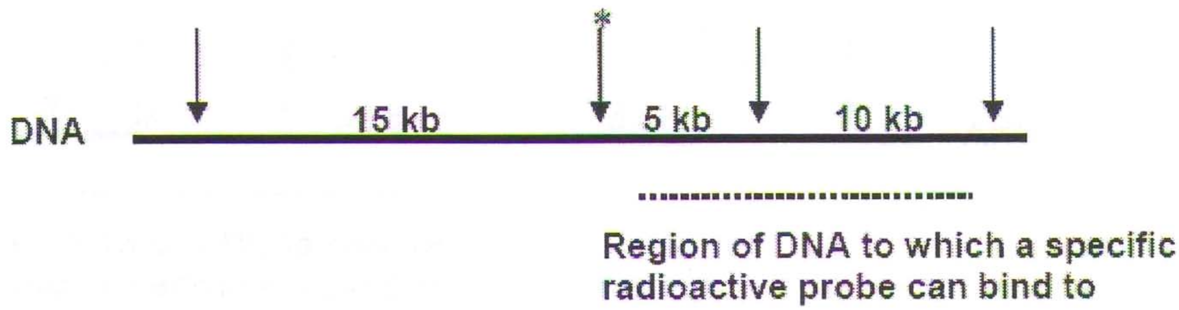


Fig. 2.1

A mutation alters one base of coding sequence at the site marked with an asterisk (*). DNA from two individuals, one who is normal and the other heterozygous for this mutation is cut with EcoR1 and the DNA fragments separated by gel electrophoresis.

a) On Fig 2.2 below,

i) Draw all the fragments of this length of DNA produced from the pair of homologous chromosomes in each of the individual tested and label the size of the fragments produced. [3]

no.; position/size of fragments; and thickness of bands;

ii) Circle the fragments which will bind to the radioactive probe and appear as black bands on an autoradiography. [2]

all correct – 2 max, 4 – 1 ½ , 3 – 1, 2 – ½ , 1 - 0

iii) Add an arrow to show the direction in which the fragments moved during electrophoresis. [1]

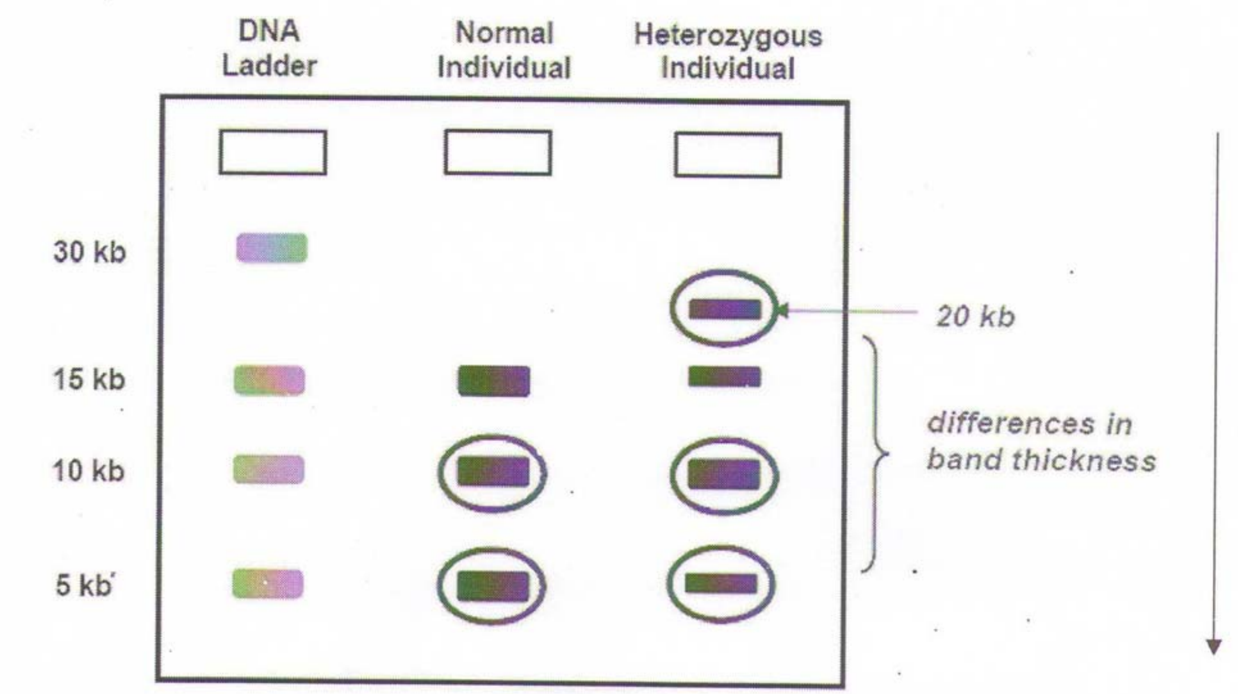


Fig. 2.2

b) State the purpose of DNA ladder. [1]

each band of known size;
serves as a basis for comparison/ reference to the size of DNA fragments/ no. of repeats present in the individuals present;

c) Explain how the DNA fragments can be separated by electrophoresis. [4]

1. **A loading dye is added to the DNA samples, which allows the process of electrophoresis to be tracked/ helps DNA to sink into the wells;**
2. **DNA samples loaded into wells at one end of an agarose gel;**
3. **Gel is submerged in a buffer solution that will conduct electricity;**
4. **An electric current is applied through electrodes at opposite ends of the gel;**
5. **Negatively-charged DNA fragments migrate to the positively charged electrode/anode;**
6. **Gel electrophoresis separates DNA fragments by size/ molecular weights;**
7. **with the smallest fragments moving the fastest and furthest;**
8. **as the smaller fragments have lower resistance in moving through the pores of the gel;**
9. **whereas the larger fragments move slower due to higher resistance;**
10. **invisible DNA bands can be viewed under UV light by staining the gel with a dye such as ethidium bromide;**

d) Three male whales have wandered into a marine research facility and fathered a calf. In order to find out which of the males have fathered the new-born calf, scientist extracted DNA from the blood samples of the mother, calf and the three males and carried out a test.

	Locus on Chromosome				
	D3S1358	vWA	FGA	D8S1179	D21S11
Mother	16, 16	15, 16	20, 22	13, 14	28, 30
Calf	16, 18	16, 16	22, 24	13, 14	28, 29
Male 1	16, 18	16, 16	22, 23	14, 15	28, 31
Male 2	17, 18	15, 16	24, 25	11, 12	28, 30
Male 3	15, 18	16, 16	19, 24	12, 13	29, 31

i) State the name of the test that was carried out. [1]

paternity testing/ DNA fingerprinting

ii) Which male, if any, may possibly be the father of the calf? [1]

male 3;

iii) The type of test performed above can only establish that a male is likely to be the father of a given offspring to a given probability. Explain why it cannot confirm that a given male is the father with absolute certainty. [2]

1. **only a limited number of loci were considered for the test;**
2. **it is possible that a male may be a match at all the loci considered;**
3. **but not be a match at another loci that was not considered in the test;**
4. **we cannot say that a male is a match with absolute certainty unless every STR loci in the whole genome is considered;**

[total: 15]

Question 3

Cystic Fibrosis (CF) is one of the most common lethal inherited diseases in humans. Worldwide there are approximately 50,000 sufferers, with an additional 10 million carriers. CF is a result of a single gene mutation and demonstrates autosomal, recessive inheritance.

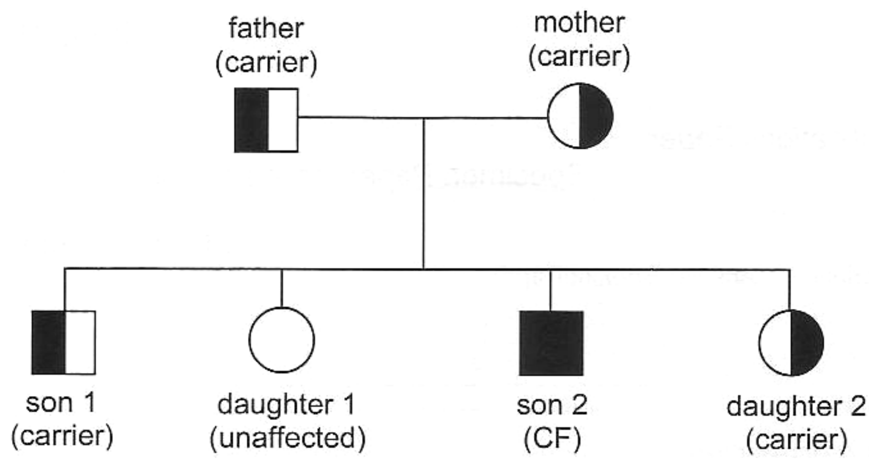


Fig. 3.1

Figure 3.1 shows the inheritance of the disease in a family in which both parents are carriers.

a) Explain how the diagram in Figure 3.1 shows that the disease is

(i) recessive [2]

disease does not show when only one copy of the allele is present/ disease only show when both copies of alleles are present;
only homozygote exhibit CF;
children can have CF when neither parent has CF;
eg: son 1 has 1 CF allele and it is only a carrier;
while son 2 has 2 CF alleles and is a sufferer;

(ii) autosomal [2]

autosome = non-sex chromosomes;
autosomal means the genes cannot be carried on sex chromosomes/
disease is independent on sex;
if disease is sex-linked, then father cannot be carrier/ son from diseased mother must be sufferer;

The CF gene is found on chromosome 7 and in non-sufferers it gives rise to a protein called Cystic Fibrosis Transmembrane Conductance Regulator (CFTR). The defect which most commonly leads to the disease involves the deletion of three nucleotides from the gene. This is illustrated in Figure 3.2.

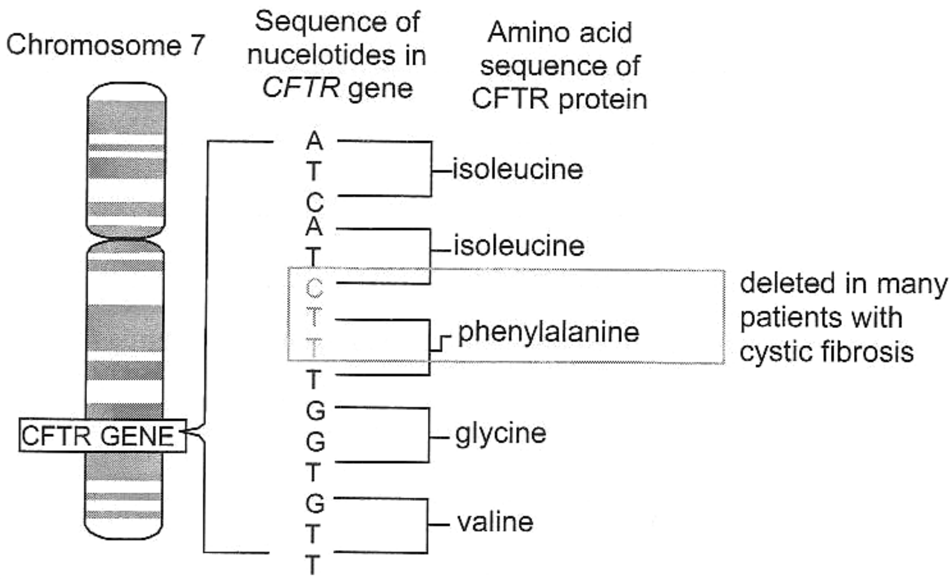


Fig 3.2

b) With reference to Figure 3.2, explain how this deletion results in the production of a defective protein, leading to cystic fibrosis. [3]

1. **3 nucleotides are removed disrupts two adjacent triplet codes;**
2. **ATCTTT replaced with ATT;**
3. **phenylalanine no longer coded for;**
4. **A form of frame-shift mutation;**
5. **ATT is a stop triplet code;**
6. **Early termination of translation of mRNA;**
7. **Results in a shorter polypeptide;**
8. **Primary structure is now different leading to defective protein formed;**
9. **CFTR protein now unable to transport chloride ions across membrane, leading to mucus buildup (AVP about CF);**

One type of treatment for cystic fibrosis involves gene therapy, using liposome as a vector. Figure 3.3 below shows the 2 routes possible for delivery of normal gene into a cystic fibrosis patient via liposomes.

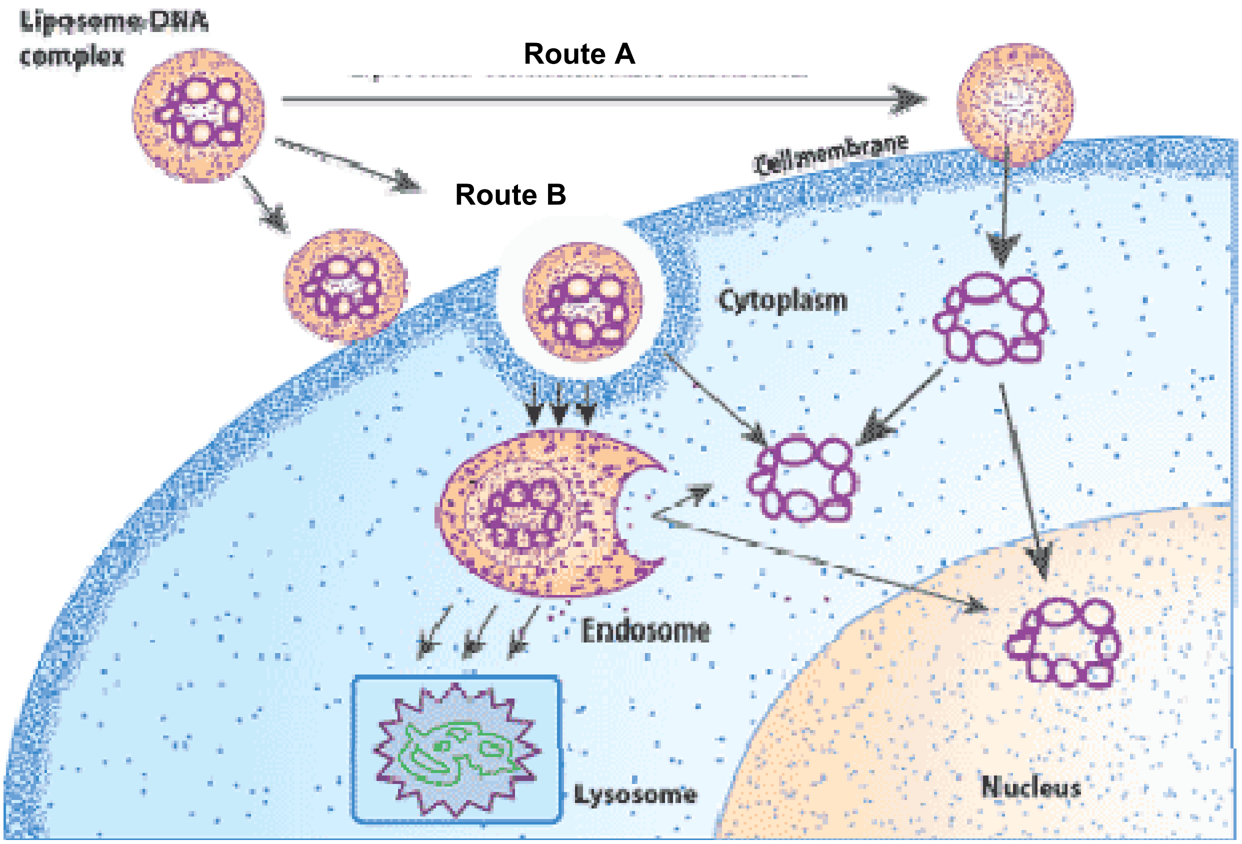


Fig. 3.3

c) With reference to Fig. 3.3, describe the 2 possible routes of gene delivery into cells using liposomes during gene therapy. [4]

Route A: liposome fuses with the cell membrane;
Fluid property of liposome membrane allows it to fuse with cell membrane which has the same fluid property/both membrane of liposome and cell is made of phospholipids;
Healthy gene enters into the cytoplasm of cell;
And may enter the nucleus as an extra chromosome;

Route B: liposome enters via endocytosis;
Invagination of cell membrane to surround liposome;
Sealing off the neck of flask/ invagination;
Endosome is then formed in the cytoplasm;
Lysosome then breaks down the two layers of membrane to release healthy genes;

Besides using liposomes for gene delivery, viral methods have also been used.

d) State 2 different *types* of viral vectors, and describe one advantage and one disadvantage of each.
[4]

Viral type : **retrovirus**

Advantage: **integrates healthy gene into genome;**
Permanent;
OR
Infects dividing cells;
Thus can be used for targeting cancer treatment;

Disadvantage: **may result in insertional mutation;**
Increasing the chance of cancer;

Viral type: **adenovirus**.

Advantage: **viral genome does not integrate into chromosome;**
Thus eliminate the problem of insertional mutagenesis by the vector;
OR
Broad host range;
Can be used to infect a large number of different types of human cells;

Disadvantage: **healthy gene inserted does not integrate into genome;**
Therapy is not permanent/ requires multiple treatments;

Viral type: **adeno-associated virus (AAV)**.

Advantage: **viral genome integrates into chromosome at specific site;**
Permanent;
eliminate the problem of insertional mutagenesis by the vector;

Disadvantage: **can carry only small DNA sequences, restricting the type of disease for treatment.**

[15 marks]

a) Explain the advantages and limitations of polymerase chain reaction (PCR). [6]

Advantages:

Speed and ease of use

1. **Takes only a few hours to run 30 cycles of PCR/a few hours for billions of copies of DNA;**
2. **Compared to a few weeks using bacteria;**
3. **The heat stable *Taq* DNA polymerase allows automation. (No need to change enzymes at each cycle);**

Sensitivity

4. **PCR can amplify minute amount of DNA, even from a single cell;**
5. **Especially useful in forensic work where DNA may not be present in high quantity;**

Specific

6. **PCR is very specific of the sequence of DNA to be amplified due to the specific primers used;**

Robustness

7. **PCR allow amplification of specific sequences from material in which the DNA is partially/badly degraded or embedded in a medium from which conventional DNA isolation is problematic;**
8. **Thus suitable for DNA studies of fossils or in formalin-preserved tissue samples;**

Limitations

need information on the target DNA sequence;

9. **since primers flanking the DNA sequence has to be manufactured;**
10. **the sequence of DNA is thus required, especially at the <u>flanking / sides regions</u>;**

short size of PCR products

11. **most PCR products are short (about 0.1 to 5kb only);**
12. **as compared to cell based cloning which can be up to 20 kb;**

limiting amount of PCR product

13. **amount of PCR product obtained in a single reaction is also much more limited;**
14. **than the amount that can be obtained using cell-based cloning where scale-up of the volumes of cell cultures is possible.**

infidelity of PCR

15. **mutation rate of the PCR product is high;**
16. **as the most widely used is *Taq* DNA polymerase which has no associated 3’ → 5’ exonuclease to confer a <u>proofreading function</u>;**

b) Explain the functions of stem cells in a living organism, using appropriate examples to illustrate. [8]

1. definition: Stem cells are cells of the body whose function is not yet determined and has the potential to differential into different cell types;

2. There are 2 main types – embryonic stem cells and adult stem cells;

3. Embryonic stem cells – can be classified as <u>early embryonic</u> and <u>blastocyst embryonic</u> stem cells;

4. Early embryonic stem cells – morula – <u>totipotent;</u>
5. Function – to differentiate into all cell types of an individual;

6. Blastocyst embryonic stem cells;
7. Obtained from inner cell mass;
8. <u>Pluripotent</u>;
9. ability to become almost any kind of cell in the body;
10. Functions: develop into embryo;

11. fetal stem cells
12. pluripotent – develop into all tissues before birth;

13. umbilical cord stem cells;
14. develop into a limited range of cell types in the umbilical cord;

15. Adult stem cells (somatic stem cells) – undifferentiated cell found among differentiated cells in a tissue or organ;
16. The primary roles of adult stem cells in a living organism are to maintain and repair the tissue in which they are found;
17. <u>multipotent</u>

18. Eg: Hematopoietic stem cells give rise to all the types of blood cells: red blood cells, B lymphocytes, T lymphocytes, natural killer cells, neutrophils, basophils, eosinophils, monocytes, macrophages, and platelets;

19. can also be <u>unipotent</u>;
20. epidermal stem cells give rise to skin cells;

c) Discuss the advantages and disadvantages of micropropagation/plant tissue culture. [6]

<u>Advantages</u>

Rapid multiplication.

1. **When shoots are produced during cloning, they produce buds which can be used to generate more shoots using the same tissue culture technique. By constantly recycling buds in this way, the number of plants produced is multiplied at each stage;**

2. **Plants with desired traits can be multiplied rapidly than can be done using conventional breeding methods, which rely on sexual reproduction.**

Genetic uniformity

3. **Since plants produced are <u>genetically identical</u>, they <u>all possess the desirable features</u> of the stock plants. (It is difficult to produce plants that breed true (homozygous for desired traits) when sexual reproduction is used).**

Production of disease-free plant material

4. As viruses do not normally penetrate to the tips of meristems. It is therefore possible to use meristems for producing virus-free plants by tissue culture.

5. Since the plants are prepared in sterile conditions they are also free from surface bacteria and fungi, some of which would cause disease.

Take up little space

6. Compared with plants growing in fields, tissue culture takes up relatively little space

Ability to air freight large quantities

7. Air freighting is cheaper because the cultures are not bulky.

Independent of climate changes

8. Tissue culture can be carried out independently of seasonal changes in climate.
9. Plants can be produced out of season, when they can be sold for higher prices.

Effective means of asexual reproduction in some plant species

10. Some plants like banana are sterile and have to be propagated asexually.

11. Some plants like orchids have seeds that are difficult to germinate. These plants can be produced more reliably by asexual means.

Genetic engineering

12. Tissue culture can be linked to genetic engineering to produce transgenic plants with improved traits;

Disadvantages

Difficulty in maintaining sterile environment

13. The greatest problem faced by commercial tissue culture laboratories is contamination. Contamination can cause very high losses in a short time. (Many plant species are easily infected by bacteria, even in very clean environments).

High production costs

14. Micropropagation requires sophisticated facilities, sterile laboratory conditions and special nutrient media. It also requires trained personnel with specialized skills.

Labour intensive

15. A lot of labour is required to transfer the plantlets from the laboratory to the field.

Higher than acceptable levels of somatic variation.

16. Callus cultures sometimes undergo genetic changes. Only some of these changes are commercially useful

Vulnerable to diseases

17. Since the clones are genetically identical, crops are very susceptible to new diseases or changes in environment. Whole crops could be wiped out.